ThermoScope Pure-fluid thermodynamic simulator

Prediction of PVT behaviour, enthalpy, entropy and fugacity using four cubic equations of state.

1. Project brief and software design

ThermoScope evaluates the thermodynamic state of a pure fluid from temperature and pressure. It compares van der Waals (vdW), Redlich-Kwong (RK), Soave-Redlich-Kwong (SRK) and Peng-Robinson (PR) predictions, with explicit treatment of volume roots and reference states. The supported components are methane, carbon dioxide, nitrogen and oxygen.

The delivered program is a single Python file containing a fluid database, numerical engine, local HTTP server and browser interface. It requires Python 3.10 or later and no third-party packages at runtime. All thermodynamic calculations are performed in Python; JavaScript presents the results and draws the plots. Input pressure can be entered in bar or MPa, while the engine uses Pa and m^3/mol internally.

Working Prototype

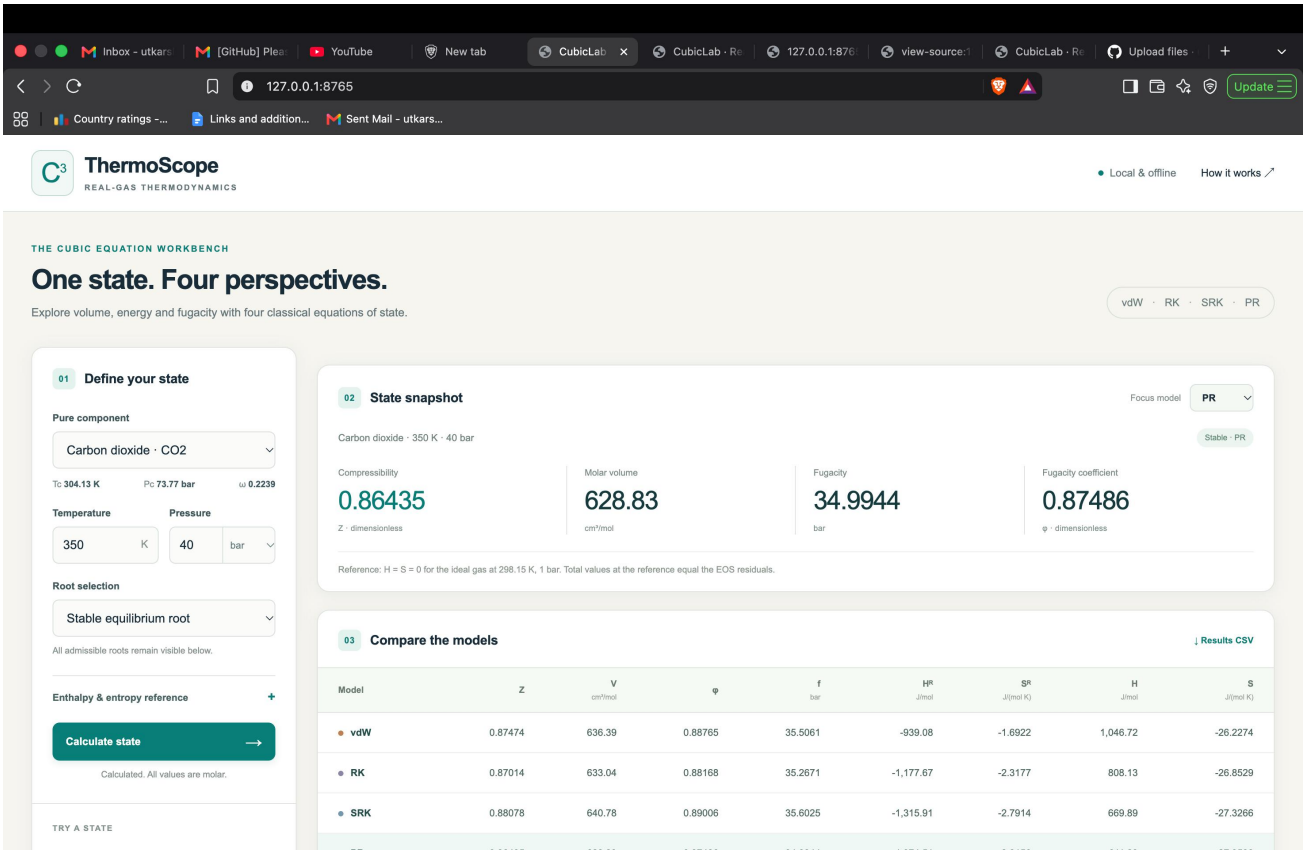


Figure 1. User-supplied screenshot of the ThermoScope prototype. The selected PR state is CO2 at 350 K and 40 bar: Z = 0.86435, v = 628.83 cm^3/mol and fugacity = 34.9944 bar. These values agree with the packaged engine. The screenshot displays the prototype at local port 8765; the single-file package defaults to port 8770.

The interface includes a model comparison table, reference-state controls, root diagnostics, pressure sweeps, raw P-v isotherms, CSV downloads and SVG plot export. A focus-model selector highlights the chosen EOS without discarding the other predictions.

Cubic EOS and root analysis

2. Unified pressure model

```
P = RT/(v-b) - a(T)/[(v+delta1*b)(v+delta2*b)]
a(T) = Omega_a R^2 Tc^2 alpha(T)/Pc
b = Omega_b R Tc/Pc; Tr = T/Tc
```

Here v denotes molar volume and Tc, Pc and omega are the critical temperature, critical pressure and acentric factor. R = 8.31446261815324 J/(mol K). The generalized volume offsets distinguish the four models [1].

Model	delta1	delta2	Omega_a	Omega_b	alpha
vdW	0	0	27/64	1/8	1
RK	1	0	0.42748	0.08664	Tr^(-1/2)
SRK	1	0	0.42748	0.08664	[1+k(1-sqrt(Tr))]^2
PR	1+sqrt(2)	1-sqrt(2)	0.45724	0.07780	[1+k(1-sqrt(Tr))]^2

```
SRK: k = 0.480 + 1.574*omega - 0.176*omega^2
PR: k = 0.37464 + 1.54226*omega - 0.26992*omega^2
```

3. Numerical solution and phase selection

With u = delta1 + delta2, w = delta1 delta2, A = aP/(RT)^2, B = bP/(RT) and Z = Pv/(RT), the pressure equation becomes:

```
Z^3 + [(u-1)B-1]Z^2
+ [A-uB+(w-u)B^2]Z - AB - wB^2(1+B) = 0
```

The custom solver bounds the roots and divides the polynomial at its stationary points. It bisects each sign-changing interval for up to 180 iterations and checks stationary points for repeated roots using a floating-point error scale. Mathematical roots are reported separately from admissible fluid roots, which must satisfy Z > B.

Mechanical stability is determined from $(dP/dv)_T$. The implementation classifies a root as marginal when the magnitude of the dimensionless slope $(dP/dv)v^2/(RT)$ is below 10^{-8}; other roots are stable for negative slope and unstable for positive slope. Equilibrium selection minimizes ln(phi) among stable or marginal candidates. Forced outer-root selections retain metastability labels.

4. Residual properties and fugacity

For J = integral from v to infinity of dx/[(x+delta1 b)(x+delta2 b)], the analytical departures from an ideal gas at the same T and P are [2]:

```
J = ln[(v+delta1*b)/(v+delta2*b)]/[b(delta1-delta2)]
vdW limit: J = 1/v; a_T = da/dT
H^R = RT(Z-1) + (T*a_T-a)J
S^R = R ln(Z-B) + a_T J
ln(phi) = Z-1-ln(Z-B)-aJ/(RT); f = phi*P
```

The derivative a_T is evaluated analytically for each EOS. A log1p form of J reduces cancellation in dilute states. The identity $H^R - TS^R$ = RT ln(phi) is checked during validation.

Heat capacities and reference states

5. Temperature-dependent ideal-gas contribution

The ideal-gas heat capacity uses piecewise Shomate coefficients A-E from the supplied NIST WebBook-based database [3]. With t = T/1000, integration gives the following primitives; differences are summed over every temperature segment crossed:

```
Cp(T) = A + B*t + C*t^2 + D*t^3 + E/t^2
F_H(T) = 1000[A*t + B*t^2/2 + C*t^3/3 + D*t^4/4 - E/t]
F_S(T) = A*ln(t) + B*t + C*t^2/2 + D*t^3/3 - E/(2*t^2)
H^ig = integral(Tref to T) Cp dT
S^ig = integral(Tref to T) Cp/T dT - R ln(P/Pref)
```

H is expressed in J/mol and S in J/(mol K). Formation enthalpies and absolute standard entropies are not imported. Piecewise definite integration avoids relying on separately rounded integration constants.

Fluid	Supported Cp intervals / K
Methane	298-1300; 1300-6000
Carbon dioxide	298-1200; 1200-6000
Nitrogen	100-500; 500-2000; 2000-6000
Oxygen	100-700; 700-2000; 2000-6000

If either the state temperature or reference temperature lies outside the supported intervals, total H and S are unavailable and a warning is shown. EOS residual properties are still returned. This is particularly relevant to the methane preset at 150 K.

6. Two explicit reference conventions

Ideal-gas zero: $H = H^{ig} + H^{R}$ and $S = S^{ig} + S^{R}$, with the ideal terms zero at the reference. **Real-fluid zero:** the stable reference residual H^{R}_{ref} and S^{R}_{ref} are also subtracted, separately for each EOS. Defaults are 298.15 K and 1 bar. Negative relative values are permissible.

7. CO2 example: 350 K, 40 bar

The table uses the default ideal-gas reference and is generated from the packaged program. All four models have one admissible root at this state.

EOS	Z	v / cm3 mol-1	f / bar	H / J mol-1	S / J mol-1 K-1
vdW	0.87474	636.39	35.5061	1046.72	-26.2274
RK	0.87014	633.04	35.2671	808.13	-26.8529
SRK	0.88078	640.78	35.6025	669.89	-27.3266
PR	0.86435	628.83	34.9944	611.29	-27.3508

For PR, phi = 0.87486, H^{R} = -1374.51 J/mol and S^{R} = -2.81559 J/(mol K). Adding the integrated ideal terms, 1985.80 J/mol and -24.53521 J/(mol K), yields H = 611.29 J/mol and S = -27.35081 J/(mol K).

Validation and practical limits

8. Reproducible checks

The single-file package provides eight executable checks. They were rerun successfully for this report. These checks demonstrate numerical and thermodynamic consistency; they do not establish agreement with experimental measurements.

Check	Scope and acceptance
Pressure closure	Four fluids, four EOS and three T,P pairs; each admissible root reconstructs pressure within 1e-6 relative error.
Gibbs identity	CO2 at 350 K and 40 bar: H^R - TS^R - RT ln(phi) within 1e-8 J/mol.
Ideal-gas limit	Nitrogen at 300 K and 1e-5 bar: both Z and phi approach 1 within 1e-6.
Real-fluid reference zero	At 298.15 K and 1 bar, selected real-reference H and S vanish within 1e-8 in their respective units.
Piecewise Cp integration	Central-difference derivatives of ideal H and S match Cp and Cp/T at four selected temperatures.
Three-root selection	Methane PR at 150 K and 10 bar: three roots, unstable middle root, minimum-ln(phi) selection.
Cp range handling	Methane at 150 K retains residuals, warns about data coverage and returns unavailable total H and S.
Fugacity derivative	CO2: H^R agrees with -RT^2(d ln(phi)/dT)P within 1e-4 J/mol.

Browser checks also confirmed initial four-model calculation, plot generation, the validation display, the methane preset and CSV download. The fluid data and browser interface were supplied with the project; the packaging added the local HTTP adapter and executable checks.

9. Scope, limitations and execution

Accepted EOS inputs span 40-6000 K and 10^{-8}-5000 bar. These are software bounds, not accuracy limits. The program warns about states far from the critical-property fitting region, below the triple point and near critical conditions. Classical cubic equations do not model solid phases or determine global phase stability against solids. Fluid roots labelled marginal require particular caution.

Phase fractions, mixtures and a saturation-pressure solver are outside scope. Roots within 10^{-7} in ln(phi) of the minimum are treated as tied; at coexistence, endpoint properties alone do not fix bulk two-phase H and S. Raw isotherm loops are mathematical homogeneous-fluid curves, not equilibrium plateaus. Pressure sweeps mark detected branch changes as plotting gaps.

Run **python3 CubicLab_Friend.py** and open **http://127.0.0.1:8770**; Windows users can use **py**. Run **python3 CubicLab_Friend.py --test** to repeat the checks. The interface can also run validation. All calculations and assets are available offline.

References and data provenance

[1] NIST teqp, General cubics: https://pages.nist.gov/teqp-docs/en/latest/models/cubics.html
[2] IDAES, Cubic Equations of State: https://idaes-pse.readthedocs.io/en/2.11.0rc1/explanations/components/property_package/general/eos/cubic.html
[3] NIST Chemistry WebBook, SRD 69, gas-phase Shomate tables for CH4, CO2, N2 and O2; Chase (1998): https://webbook.nist.gov/chemistry/
[4] CoolProp primary fluid data, critical constants and acentric factors: https://github.com/CoolProp/CoolProp/tree/master/dev/fluids
The supplied database records retrieval on 10 November 2026. Figure 1 was provided by the user.